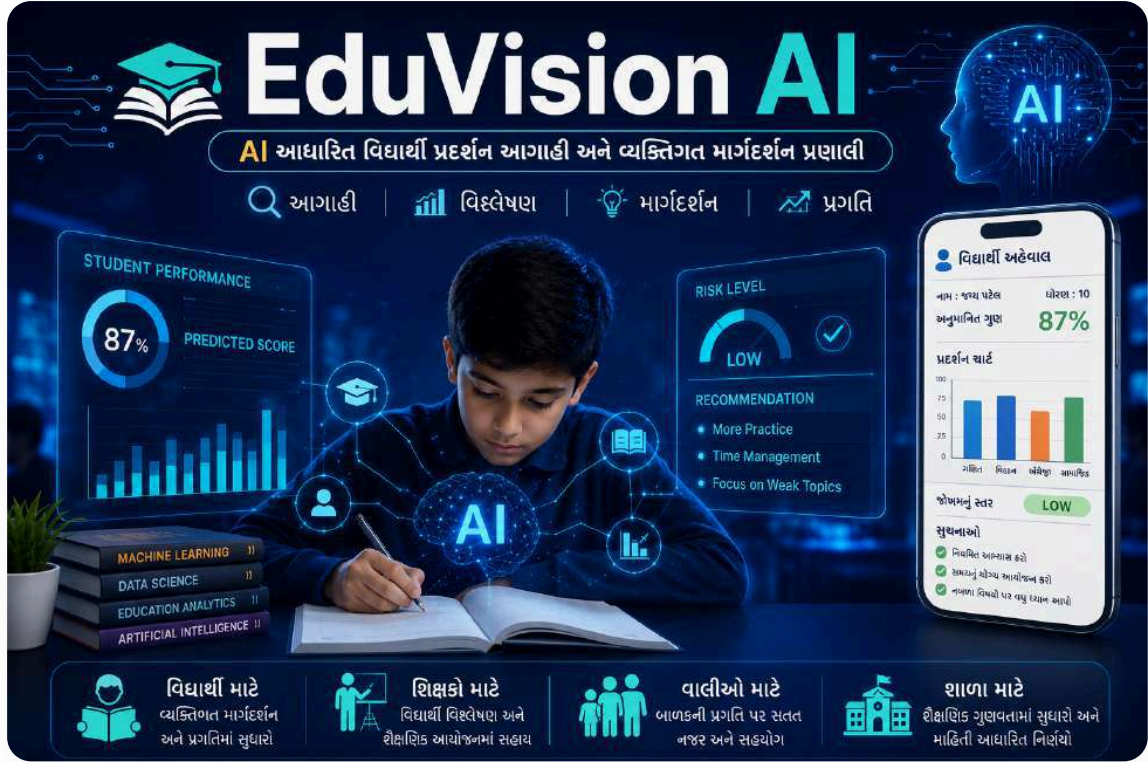




EduVision AI

AI-Powered Student Academic Performance Prediction & Personalized Mentorship System

Project Live Website: <https://eduvision-ai-kappa.vercel.app/>
(Alternative: <https://eduvision-ai.streamlit.app/>)

Project Mentor: Shri Manojbhai Parmar

School: M. M. Karodiya Primary School, Tarsadi Kosamba (R.S)

Developer & Researcher: Yash Patel (Science Fair 2026)

Project Title

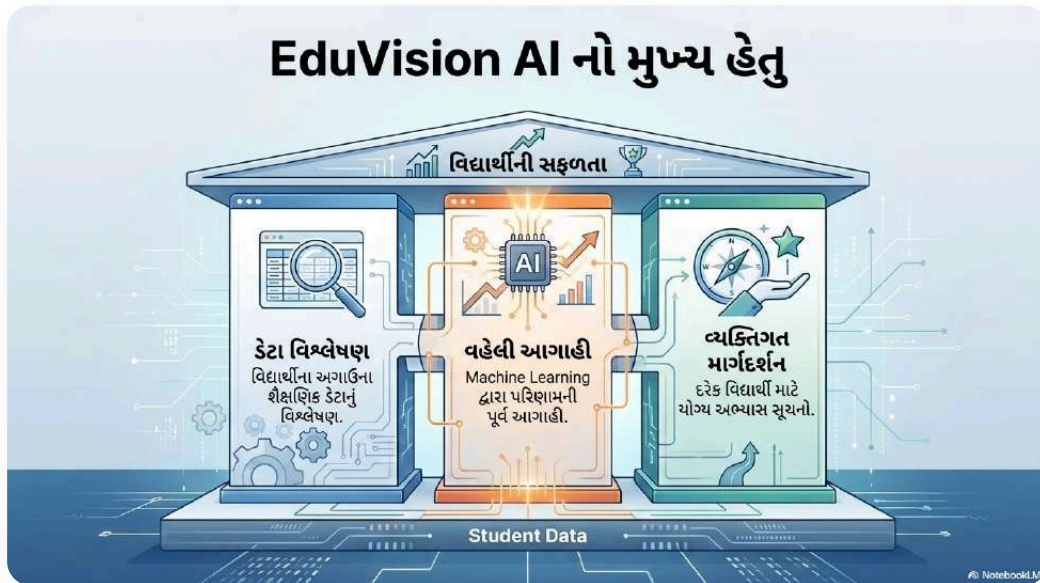
EduVision AI - Artificial Intelligence Powered Student Performance Prediction and Personalized Academic Mentorship System.

A professional, production-grade educational intelligence dashboard predicting final grades, assessing academic risk levels, and generating tailored study recommendations based on multi-dimensional student data.

Project Objectives

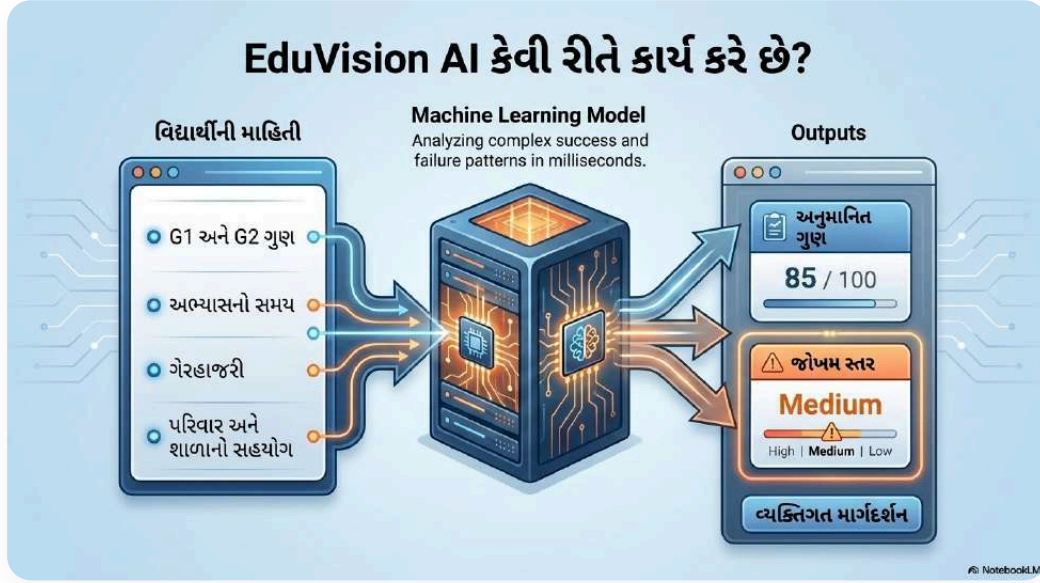
The primary objective of this project is to leverage cutting-edge **Artificial Intelligence (AI)** and **Machine Learning (ML)** technologies to analyze students' historical academic marks, daily study routines, behavioral factors, and attendance records, thereby accurately forecasting future academic performance and providing proactive, timely mentorship.

The system enables educators to identify academically vulnerable students well in advance of final board examinations, facilitating targeted remedial intervention. Simultaneously, it empowers students with clear self-awareness of their learning habits, fostering measurable performance improvement and academic confidence.



Core Scientific Principle

A student's future academic performance is strongly correlated with preceding internal evaluations (G1 and G2 exam scores) along with behavioral variables (daily study duration, school absences, educational support, and health status). By applying non-linear ensemble regression algorithms, these interactions can be mapped precisely, enabling timely corrective interventions that maximize academic potential.



System Specifications & Requirements

Hardware Requirements

- Modern Computing Device (Laptop / Desktop / Tablet) with High-Speed Internet Connection

Software & Development Environment

- Python 3.11+ Runtime & Next.js 14 Production Web Application
- FastAPI High-Performance Asynchronous Python Microframework
- Visual Studio Code (VS Code) & Jupyter Notebook Development Suite
- Anaconda Package & Virtual Environment Manager

Python Scientific Libraries

- **Pandas:** High-performance structured data manipulation, CSV processing, and tabular analysis.
- **NumPy:** Numerical linear algebra computations, array transforms, and statistical calculations.
- **Scikit-Learn:** State-of-the-art machine learning model training, pipelines, and evaluation metrics.
- **Matplotlib & Seaborn:** Statistical data visualization, distribution plots, and correlation heatmaps.
- **Joblib:** High-efficiency serialization and persistent storage of the trained machine learning pipeline.
- **FastAPI & Uvicorn:** Production REST API server delivering low-latency inference endpoints.

Dataset Benchmark Information

- **Benchmark Dataset:** Student Performance Dataset
- **Repository Link:** <https://archive.ics.uci.edu/dataset/320/student+performance>
- **Dataset Description:** Curated from the UCI Machine Learning Repository, comprising 395 verified student records spanning 33 demographic, academic, and social variables, capturing comprehensive behavioral profiles and subsequent academic outcomes.

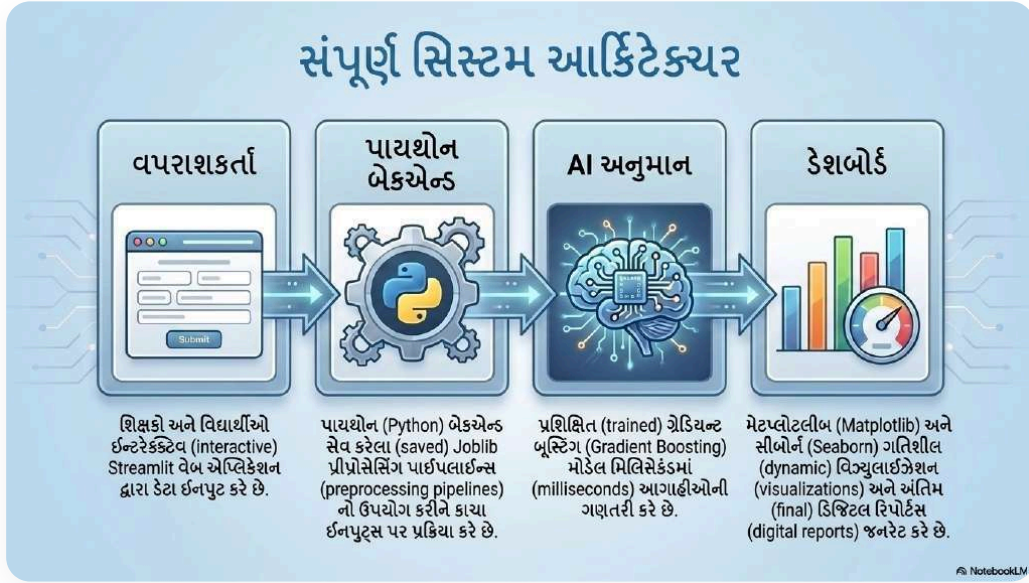
System Architecture

EduVision AI is engineered as a robust, decoupled, cloud-ready web architecture. The framework comprises five foundational tiers: User Interaction Tier, Responsive Web Application (Next.js 14), Feature Preprocessing Pipeline, Machine Learning Inference Engine (Gradient Boosting), and Personalized Output Generation Module.

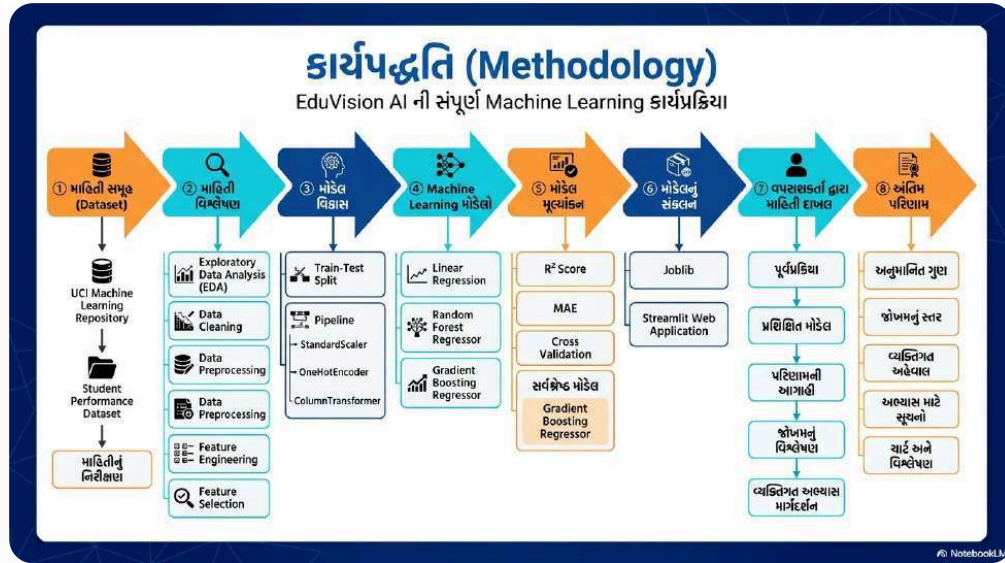
The user inputs student credentials and evaluation parameters through a responsive interface. The incoming payload undergoes real-time schema validation and transformation within the preprocessing pipeline.

The normalized features are forwarded to the trained Gradient Boosting model, which computes an exact numerical prediction of the student's expected final score (G3) out of 20.

Based on the prediction and key feature importances, the system categorizes the student into precise risk bands (Excellent, Good, Moderate, or High Attention) and generates dynamic remedial advice. Finally, the student report card and interactive dashboard visualizations are rendered instantaneously.



Machine Learning Methodology

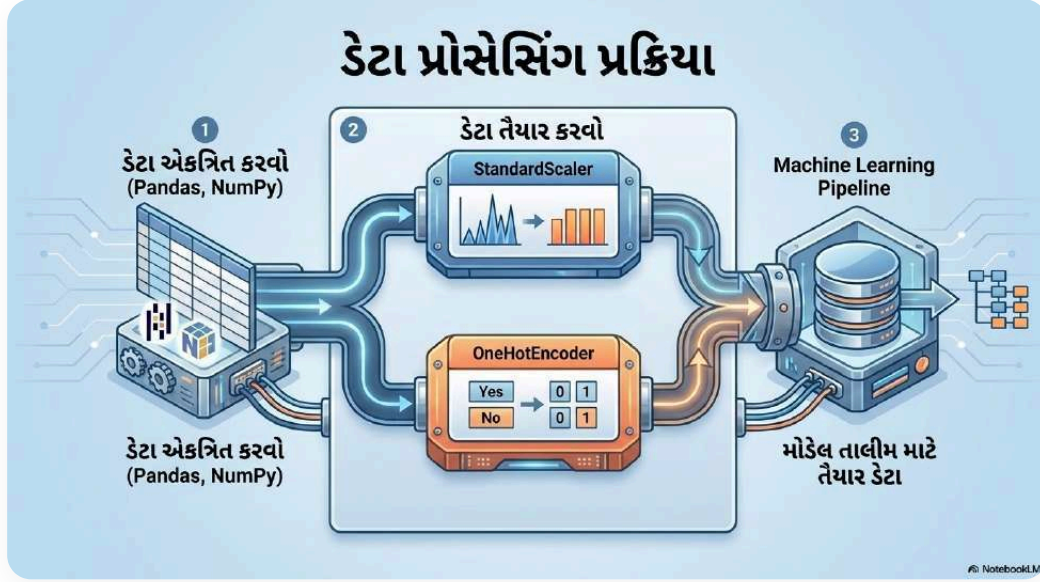


The development methodology follows a rigorous 7-stage Machine Learning lifecycle. Beginning with dataset acquisition from the UCI Machine Learning Repository, comprehensive quality verification and missing-value diagnostics were conducted.

Subsequent Exploratory Data Analysis (EDA) mapped the mathematical correlations between study hours, absenteeism, and evaluation scores. Rigorous data cleaning, feature engineering, and feature selection isolated the 12 most impactful academic indicators.

The dataset was partitioned into an 80% training set and a 20% independent testing set. Feature preprocessing pipelines incorporating numerical standardization (StandardScaler) and categorical encoding (OneHotEncoder) were integrated using Scikit-Learn's ColumnTransformer...

The unified ColumnTransformer guarantees deterministic, identical preprocessing transformations across both training and real-time production inference.



Candidate regression algorithms—including Linear Regression, Random Forest Regressor, and Gradient Boosting Regressor—were rigorously trained and validated. Comprehensive performance benchmarks evaluating the coefficient of determination (R^2 Score), Mean Absolute Error (MAE), and k-fold cross-validation confirmed that Gradient Boosting Regressor achieved the highest predictive accuracy and lowest residual variance, establishing it as the definitive production model.

Gradient Boosting Regressor શા માટે પસંદ કર્યો?

લિનિયર રીગ્રેસન	રેન્ડમ ફોરેસ્ટ	ગ્રેડિયન્ટ બૂસ્ટિંગ
MAE: 1.4429	MAE: 1.2061	MAE: 1.1804
R ² Score: 0.7690	R ² Score: 0.8132	R ² Score: 0.8138



Gradient Boosting સતત પોતાની ભૂલોમાંથી શીખે છે અને વધુ ચોક્કસ પરિણામ આપે છે.

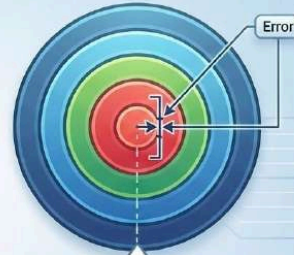
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મોડેલનું પ્રદર્શન



R² Score: 81.38%

મોડેલ લગભગ 81% સુધી સાચી આગાહી કરી શકે છે.



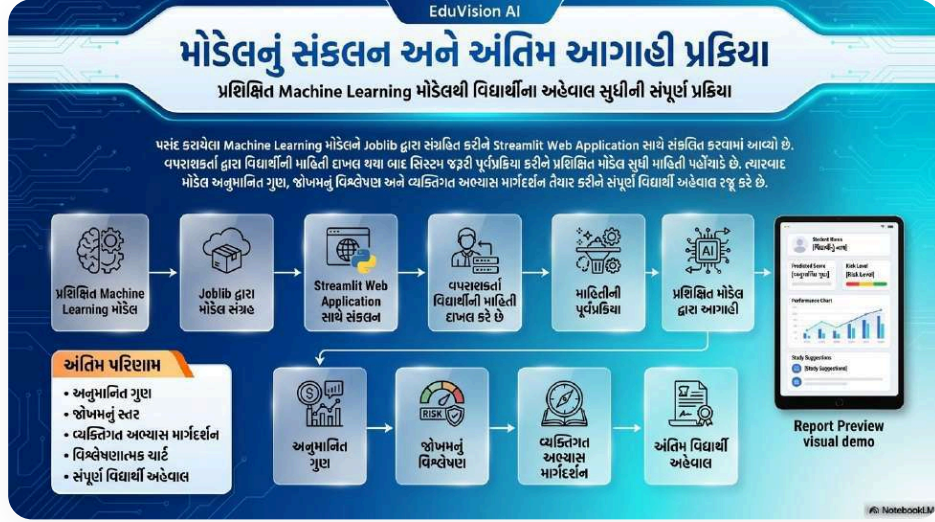
MAE: 1.18

સરેરાશ માત્ર 1 ગુણ જેટલી ભૂલ.

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The finalized model pipeline was serialized using Joblib and integrated with the production web application. When a user submits student parameters, the system executes the pre-fitted preprocessing pipeline and feeds the transformed array to the model.

The application outputs the predicted score, performance percentage, academic risk tier, and customized actionable recommendations, delivering an end-to-end intelligent decision-support system.



Applications & Stakeholder Benefits

- **Schools & Administration:** Enables data-driven strategic planning, optimized allocation of tutoring resources, and continuous monitoring of institutional academic quality.
- **Educators & Mentors:** Streamlines early detection of struggling students, reducing diagnostic overhead and facilitating timely personalized academic interventions.
- **Students:** Instills constructive self-awareness regarding study discipline and attendance, accompanied by clear, actionable pathways for grade improvement.
- **Parents & Guardians:** Provides transparent, objective insights into child academic progress, strengthening home-school collaboration.

- **Evidence-Based Decision Making:** Transforms traditional qualitative educational intuition into quantitative, evidence-based academic excellence strategies.

ઉપયોગ અને ફાયદા

EduVision AI દ્વારા શિક્ષણ ક્ષેત્રમાં મળતા મુખ્ય લાભો



① વિદ્યાર્થીઓ માટે

- ભવિષ્યના શૈક્ષણિક પરિણામની આગાહી
- નબળાઈઓની સમયસર ઓળખ
- વ્યક્તિગત અભ્યાસ માર્ગદર્શન
- અભ્યાસમાં આત્મવિશ્વાસનો વધારો



② શિક્ષકો માટે

- દરેક વિદ્યાર્થીનું સરળ વિશ્લેષણ
- નબળા વિદ્યાર્થીઓની વહેલી ઓળખ
- યોગ્ય શૈક્ષણિક આયોજન
- માહિતી આધારિત નિર્ણય લેવામાં સહાય



③ વાલીઓ માટે

- બાળકની શૈક્ષણિક પ્રગતિની માહિતી
- સમયસર યોગ્ય માર્ગદર્શન આપવામાં સહયોગ
- બાળકના અભ્યાસ પર સતત નજર રાખવાની સુવિધા



④ શાળા માટે

- શૈક્ષણિક ગુણવત્તામાં સુધારો
- પરિણામનું વિશ્લેષણ સરળ બને
- શૈક્ષણિક આયોજનમાં સહાય
- Data-driven નિર્ણય લેવામાં મદદ

EduVision AI ની વિશેષતા

EduVision AI માત્ર પરિણામની આગાહી કરતું સાધન નથી, પરંતુ સમયસર યોગ્ય માર્ગદર્શન આપીને વિદ્યાર્થી, શિક્ષક, વાલી અને શાળા - ચારેયને વધુ સારા શૈક્ષણિક નિર્ણયો લેવામાં મદદરૂપ બનતી બુદ્ધિશાળી શૈક્ષણિક પ્રણાલી છે.

સમાજ અને શિક્ષણ ક્ષેત્ર પર અસર



વિદ્યાર્થીઓ માટે

મહત્વપૂર્ણ આત્મ-જાગૃતિ પ્રદાન કરે છે અને અંતિમ પરીક્ષાઓના લાંબા સમય પહેલાં અભ્યાસની ટેવો સુધારવાની અમૂલ્ય તક આપે છે.



શિક્ષકો માટે

તરત જ ધ્યાન આપવાની જરૂર હોય તેવા વિદ્યાર્થીઓની સ્માર્ટ, સ્વચાલિત વોચ-લિસ્ટ બનાવે છે, વહીવટી બોજને નોંધપાત્ર રીતે ઘટાડે છે.



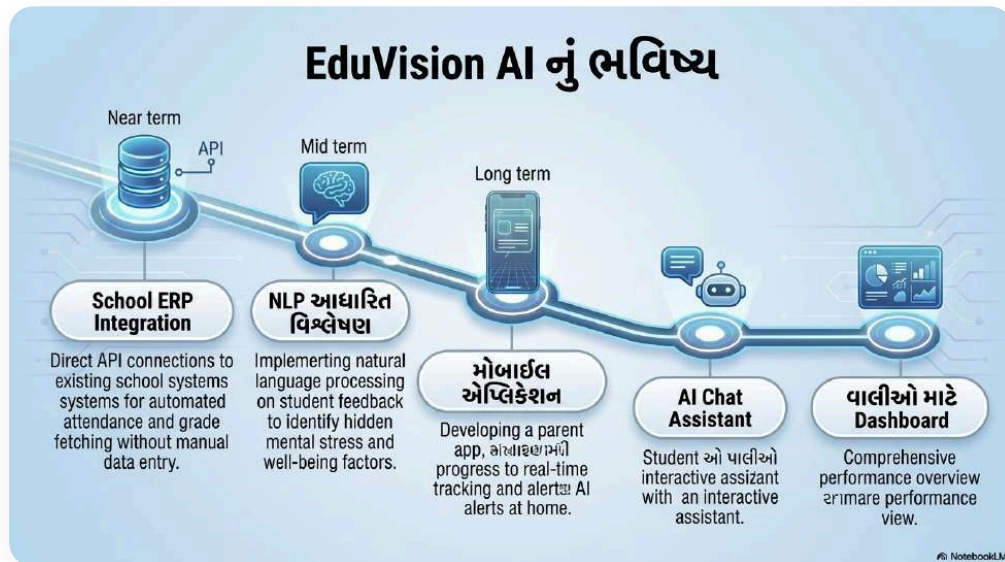
શાળાઓ માટે

ડેટા-આધારિત વહીવટ, સંસાધન ફાળવણીને શ્રેષ્ઠ બનાવે છે અને સંસ્થાકીય પાસિંગ દરમાં નોંધપાત્ર સુધારો લાવે છે.

Future Scope & Strategic Roadmap

While EduVision AI currently delivers real-time prediction and tailored academic guidance, a comprehensive multi-phase roadmap is planned to expand its capabilities:

1. **School ERP API Integration:** Direct bi-directional integration with School Management Information Systems (MIS/ERP) to automate attendance synchronization, grade logging, and student onboarding without redundant manual data entry.
2. **NLP-Based Feedback & Sentiment Analysis:** Leveraging Natural Language Processing (NLP) to analyze student qualitative feedback, teacher remarks, and emotional sentiment to identify mental stress and non-academic learning impediments.
3. **Native Mobile Applications for Parents:** Developing cross-platform iOS and Android applications providing real-time notification alerts, push updates, and weekly progress telemetry directly to parents.



4. **Interactive Conversational AI Tutor:** Implementing an AI Chat Assistant powered by Large Language Models (LLMs) to answer student curriculum queries 24/7 and deliver individualized motivational coaching.

5. **Unified Parent & School Analytics Dashboard:** Centralized portal providing aggregated longitudinal tracking, institutional performance distributions, and predictive cohort analytics.

Through these systematic expansions, EduVision AI is dedicated to pioneering an intelligent, inclusive, and student-centric educational ecosystem that empowers every learner to realize their full potential.

Conclusion & Official Endorsements

Shri Manojbhai Parmar
Project Mentor & Head
M. M. Karodiya Primary School

Yash Patel
Developer & AI Researcher
Science Fair 2026